

Minimizer Verification and DF-Freedom Degeneracy: Summary of the Latest Runs

July 12, 2026

Motivation

Before trusting the flexible-distribution-function (DF) experiments, we test the *minimizer itself*: if we hand the fit a small correction to the potential, does it still recover the dark-matter scale height h_{dm} , or does it prefer to distort the potential? Recall the physical difficulty: h_{dm} enters the potential only through anharmonic terms, so it is weakly constrained and easy to trade against other freedoms. All runs below use clean, self-generated data (static halo at $h_{\text{dm}}^{\text{true}} = 250$ pc, no satellite), so the truth is known exactly.

1 Test A — Free double-Gaussian DF + time-dependent $\delta\Phi(x, t)$

Setup. Data drawn from a genuine two-population DF (cold $\sigma_z, \sigma_w = 6, 12$; hot 25, 45; mixing $\pi = 0.6$). The fit is given *maximal* freedom: a two-component Gaussian mixture with *every* property free (both means, both dispersions of each component, and π), *plus* a time-dependent potential correction $\delta\Phi(x, t) = A_{\text{nn}} \text{MLP}(z/z_{\text{max}}, t/t_{\text{end}})$ capped at 10% of the static potential peak. Fit started off truth ($h_{\text{init}} = 200$).

Result.

parameter	truth	fitted
h_{dm} [pc]	250.0	133.0 (−46.8%)
π_{cold}	0.60	0.36
cold (σ_z, σ_w)	(6, 12)	(12.1, 5.4)
hot (σ_z, σ_w)	(25, 45)	(100.0 [†] , 31.6)

[†] pinned at the upper σ clamp. $\delta\Phi$ reached only 0.11% of the potential peak (cap 10%).

Reading. h_{dm} is badly biased (−47%), but *not* because the time-dependent correction absorbed signal — $\delta\Phi$ stayed essentially off (0.11%). The bias is driven by the **free double-Gaussian DF collapsing to a degenerate corner** (the “hot” component runs to the $\sigma_z = 100$ bound, the weight and dispersions are wrong). Handing the fit two full Gaussian components on this dataset is under-constrained: the DF freedom, not the potential, corrupts h_{dm} . This run is therefore *confounded* and cannot, by itself, tell us whether the data favor a time-dependent potential.

2 Test B — Minimizer sanity: known DF + a 10% static NN bump

Setup. Remove the DF freedom. The DF is *fixed at truth* (single Gaussian, $\sigma_z, \sigma_w = 10, 20$); only h_{dm} (started at 200) and a static NN potential correction, capped at 10% of the static potential

peak, are free. Same clean data.

Result.

parameter	truth	fitted
h_{dm} [pc]	250.0	248.75 (−0.5%)

The NN correction reached only 0.01% of the potential peak (cap 10%): it did not engage.

Reading. With the DF known, the minimizer recovers h_{dm} essentially perfectly (−0.5%) and *declines* to add a potential correction even though it is allowed up to 10%. Two conclusions:

- The optimizer is trustworthy — given the correct DF it returns the correct h_{dm} and leaves the extra potential freedom near zero.
- On this clean, static dataset the data are **rigid** to a (time-independent) potential correction: there is no feature for the NN to absorb, so it stays off.

Takeaways and next step

The contrast between A and B localises the problem: the −47% bias in Test A comes from **DF over-freedom** (an unconstrained two-Gaussian mixture), not from the potential correction, which barely moved in either run. The minimizer is sound (Test B).

Open question for the next tests. Test B used a *static* NN. To answer whether *the data favor time dependence*, the clean comparison is to repeat Test B with a *time-dependent* NN and check whether it reaches a lower NLL / moves h_{dm} relative to the static case. If both stay near 0% with $h_{\text{dm}} \approx 250$, the data are rigid to a time-dependent potential — the result we are ultimately after.